Predicting ICU Patient Deterioration and Mapping their Trajectories

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Abstract

This project explores ICU patient deterioration and mortality prediction using the eICU Collaborative Research Database. The main goal was to determine whether time-series clinical data, combined with demographic and severity information, can be used to accurately predict patient outcomes and reveal meaningful patient trajectory patterns.

The analysis followed a structured pipeline that included data preprocessing, feature engineering, exploratory data analysis, dimensionality reduction using PCA, supervised machine learning, and unsupervised clustering. Time-series ICU data such as vital signs, lab values, medications, and severity scores were aggregated into patient-level features and used to train multiple predictive models, including Logistic Regression, Random Forest, XGBoost, LightGBM, and KNN. In parallel, clustering methods (K-Means and hierarchical clustering) were applied to identify natural groupings of patients based on physiological patterns.

The results showed that gradient boosting models, particularly LightGBM, achieved strong predictive performance for hospital mortality. PCA also revealed that a small number of underlying physiological components captured most of the predictive signal, particularly those related to overall disease severity and respiratory function. Unsupervised learning further showed that ICU patients naturally form distinct subgroups associated with different levels of clinical severity and outcomes.

Overall, the findings suggest that ICU mortality is driven by complex, multi-system interactions rather than single variables, and that machine learning can effectively capture these patterns. However, limitations in interpretability, temporal modeling, and external validation highlight areas for future improvement before clinical application.

Introduction and Problem Definition

Patients admitted to the Intensive Care Unit (ICU) are among the most medically unstable individuals in the hospital. Their condition can change rapidly, and even small physiological changes can signal serious deterioration. Many of the clinical deteriorations that occur in the ICUs, such as acute respiratory failure, hemodynamic instability, or death, tend to happen after measurable physiological changes that precede these critical events by several hours. Unfortunately, these early warning signals may be subtle, nonlinear, and difficult to detect through the traditional clinical scoring systems..

This is where Machine Learning has the potential of bridging the gap. This capstone project proposes to leverage the eICU Collaborative Research Database, available through

PhysioNet, to develop a prospective analytical framework for predicting ICU patient deterioration and mapping patient trajectories over time. Specifically, this project will:

- Perform exploratory descriptive analytics to characterize ICU patient populations and visualize deterioration patterns.
- Develop predictive models for outcomes such as in-hospital mortality and need for mechanical ventilation etc.
- Apply unsupervised learning to identify clinically meaningful patient trajectory clusters.
- Potentially also explore prescriptive analytics to inform resource allocation and early intervention thresholds (I am still working on fleshing this part out a little more).

The central research question guiding this project is "Can time-series physiological data combined with demographic and severity metrics accurately predict ICU deterioration and reveal clinically meaningful trajectory patterns that support proactive decision-making?"

This project carries significant implications as early risk identification such as which patients are likely to worsen could help clinicians intervene sooner and potentially improve outcomes. It could also help optimize ICU resource allocation and enhance decision support systems in high-acuity environments. It is important to recognize that the goal is not to replace clinical judgment, but to build a data driven framework that supports earlier detection and better-informed decisions.

Literature Review

Recent research in medical outcomes has increasingly focused on applying machine learning techniques to predict patient deterioration using electronic health record (EHR) data. A particularly relevant study is the Prediction of ICU Patients' Deterioration Using Machine Learning Techniques by Aldhoayan and Aljubran (2023), which sought to evaluate the predictive power of vital sign measurements in ICU patients using multiple data mining algorithms.

In their retrospective chart review of 653 ICU patients at a hospital in Saudi Arabia, the authors applied five machine learning models: logistic regression, support vector machine (SVM), k-nearest neighbors (KNN), gradient boosting (XGBoost), and Naive Bayes. Their objective was to determine whether basic physiological indicators like blood pressure, respiratory rate, temperature, heart rate, and oxygen saturation, could predict in-hospital mortality. Using an 80/20 train-test split, they found that the gradient boosting classifier achieved the highest accuracy (88.83%), followed by KNN (84.72%). Feature ranking using SelectKBest identified diastolic blood pressure as one of the strongest predictor of deterioration, followed by systolic blood pressure and respiratory rate.

This study provides strong evidence that machine learning methods could outperform traditional statistical approaches for mortality prediction in ICU settings. One of its strengths include model comparison across multiple classifiers, systematic feature ranking, and the use of clinically interpretable variables. The findings also reinforced prior research that suggests that vital sign abnormalities can be used as early indicators of deterioration.

However, some limitations of this study highlight opportunities for expansion that this project could work on. The study relied exclusively on a small set of vital signs- about10 features. It did not incorporate laboratory values, medication administration, or severity scores such as APACHE, which are available in larger ICU datasets. The Data was also drawn from one hospital which could limit generalizability. The analysis also focused only on the first 24 hours of ICU admission and did not model longitudinal physiological trajectories. Additionally, while the study used multiple models of supervised classification, it did not explore unsupervised clustering that could have identified patient subgroups with distinct deterioration pathways.

This capstone will build directly upon and extend on their work in several ways. First, instead of relying solely on basic vital signs, this project will use the eICU Collaborative Research Database, which contains a wide range of other features such as laboratory values, medication records, mechanical ventilation status, and APACHE severity scores. This richer feature space should allow for a more comprehensive modeling. Second instead of using only static early measurements, this project will attempt to engineer time-series features that capture trends like slopes, variability and rolling averages etc. The Modeling trajectories over-time approach should also align more closely with the dynamic nature of ICU deterioration.

Third, in addition to supervised models such as logistic regression and gradient boosting, this project will incorporate unsupervised learning methods like K-Means and hierarchical clustering to identify clinically meaningful patient trajectory groups. This would address a major gap in the Aldhoayan study, which focused exclusively on classification.

In conclusion, the Aldhoayan et al. study provides a strong methodological baseline and confirms the feasibility of using machine learning to predict ICU deterioration. The proposed capstone hopes to expand the scope in terms of data richness, temporal modeling, clustering, and decision optimization.

Methodology

Hypothesis: Can we use ICU time-series data to predict patient deterioration more accurately and identify meaningful patterns in how patients worsen or recover?

Data Description

The dataset I plan to use is the eICU Collaborative Research Database, available through PhysioNet. It is a large, publicly available, de-identified ICU dataset containing patient-level information from multiple hospitals in the United States.

Preliminary Features include:

- Patient demographics (age, gender)
- Vital signs (heart rate, blood pressure, respiratory rate, oxygen saturation)
- Laboratory values
- Severity scores such as APACHE
- Mechanical ventilation status
- Mortality outcomes

This study followed a structured pipeline consisting of data preprocessing, feature engineering, exploratory data analysis (EDA), and both supervised and unsupervised modeling. The goal, as stated in the introduction, was to analyze eICU data to predict hospital mortality and identify meaningful patient subgroups.

Data Preprocessing

The analysis began with loading and cleaning multiple datasets from the eICU-CRD-demo database, including patient, admission diagnosis, lab, medication, diagnosis, treatment, and vital sign records. Each dataset was first inspected using basic functions to understand its structure, data types, and missing values.

For the patient dataset, age values recorded as "> 89" were standardized to 90 and converted to numeric, with missing ages imputed using the median. The discharge weight column was dropped due to high missingness, while admission height and weight were filled using median values. Missing categorical fields such as gender, ethnicity, and admission/discharge details were imputed using the most frequent category.

For the APACHE dataset, missing ventilation-related variables were set to 0, assuming no recorded ventilation. Mortality outcomes were converted from categorical labels (e.g., ALIVE, EXPIRED) into binary values. Remaining numerical features were imputed with 0, and categorical variables were filled with "Unknown."

In the lab dataset, rows with missing lab results were dropped since these values were not critical. The remaining lab results were then converted to numeric, and invalid entries were removed.

For the medication dataset, categorical fields such as drug name and route were filled using the mode, while numerical fields like dosage were imputed using the median. Dosage values were cleaned by extracting numeric components and converting invalid entries to 0. In the diagnosis and treatment datasets, missing text fields were replaced with "Unknown."

Lastly, the periodic vital signs dataset required additional time-series handling. Records were sorted by patient ID and time offset, and missing values were filled using forward-fill followed by backward-fill within each patient to preserve temporal consistency. Any remaining missing values were replaced with column medians, and time offsets were converted from minutes to hours for easier interpretation.

Exploratory Data Analysis

Once the dataset was clean, an exploratory data analysis (EDA) was conducted to better understand the dataset and identify patterns related to hospital mortality.

Distributions of key variables such as age and ICU length of stay were visualized using histograms with density curves. Mortality was compared across demographic variables like gender and ethnicity using count plots, and relationships between clinical severity and outcomes were explored through visualizations such as violin plots of APACHE scores by mortality and box plots of ICU length of stay.

A correlation heatmap was used to examine relationships among numerical variables, and distributions of core vital signs were visualized to understand their overall behavior. Lastly, to capture temporal patterns, time-series plots of vital signs were generated for a sample patient, showing how these measurements changed over time.

Feature Engineering

Following the EDA, feature engineering was performed, focusing on summarizing time-series and event-based data into patient-level features, particularly within the first 24 hours of ICU admission.

The lab data was aggregated by computing the mean, minimum, and maximum lab values per patient. The medication data was summarized using the number of unique medications and total dosage administered. Aperiodic vital signs were aggregated using mean, minimum, maximum, and standard deviation, while diagnosis and treatment data were transformed into counts of unique entries and key subsets (e.g., major diagnoses and treatments within 24 hours). Similarly, periodic vital signs such as heart rate, temperature, and oxygen saturation were summarized using mean, minimum, maximum, and standard deviation over the first 24 hours.

All engineered features were then merged into a single dataset using patient identifiers. Any missing values introduced during merging were imputed using medians for numerical variables and "Unknown" for categorical ones. Categorical variables were then encoded using one-hot encoding. Additionally, certain high-cardinality identifiers and time-based fields were excluded to avoid creating an excessively large feature space.

Dimensionality Reduction- PCA

After feature engineering, the dataset was prepared for modeling by separating the target variables, hospital and ICU mortality, from the input features. Features were standardized, and Principal Component Analysis (PCA) was applied to reduce dimensionality while retaining 95% of the variance. The resulting components were then used as inputs for modeling

Supervised Predictive Modeling

Data Splitting

The dataset was split into training (80%) and testing (20%) sets, with stratified sampling was used to preserve the distribution of hospital mortality due to class imbalance.

Supervised Learning Models

Several models were trained to predict hospital mortality:

- Logistic Regression was used as a baseline model with balanced class weights.
- Random Forest was trained with class balancing to handle imbalance.
- LightGBM and XGBoost were used as gradient boosting approaches, with built-in mechanisms to address imbalance.
- K-Nearest Neighbors (KNN) was also included as a non-parametric method.

LightGBM was then further optimized using randomized hyperparameter search with cross-validation, focusing on ROC-AUC as the evaluation metric.

Unsupervised Learning

Lastly, in order to identify patient subgroups, unsupervised learning was applied to aggregated vital sign features.

A separate feature set was created by computing statistical summaries including mean, median, standard deviation, minimum, and maximum of key vital signs. These features were then standardized prior to clustering.

Two clustering approaches were used. First, K-Means clustering was applied, with the number of clusters selected using the Elbow Method and Silhouette Score. Patients were then assigned to clusters, and each cluster was analyzed based on clinical and demographic characteristics, including mortality, age, and ventilation days.

Second, hierarchical clustering was performed using Ward's method. A dendrogram was used to visualize the clustering structure, and clusters were defined based on a selected threshold. These clusters were then analyzed in the same way as the K-Means results.

Results and Analysis

Exploratory Data Analysis Findings

The exploratory data analysis (EDA) provided an initial understanding of the patient population, outcome distributions, and relationships between key clinical variables.

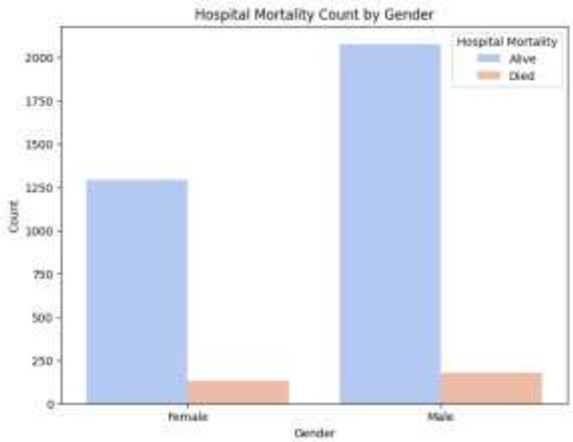

Figure 1.1 Hospitality Mortality By Gender

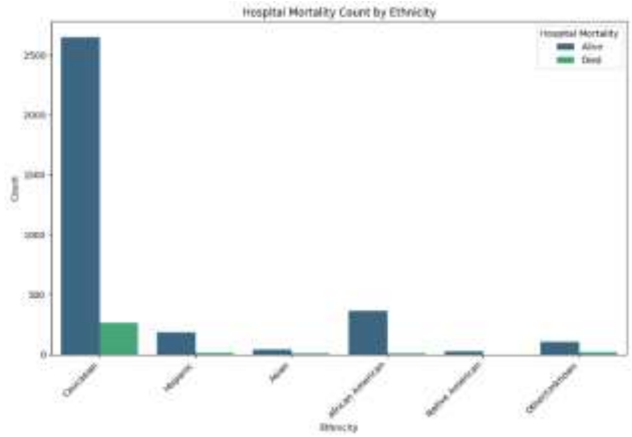

Figure 1.2 Hospitality Mortality By Ethnicity

The distribution of hospital mortality across gender showed that although there were more male patients overall, the proportion of deaths was relatively similar between genders. This suggests that gender alone may not be a strong standalone predictor of hospital mortality, but it could still contribute in combination with other variables.

In contrast, mortality across different ethnic groups showed variation in raw counts, but these differences alone do not directly indicate risk without normalization. This reinforces the need to include ethnicity as a feature rather than drawing conclusions from raw distributions..

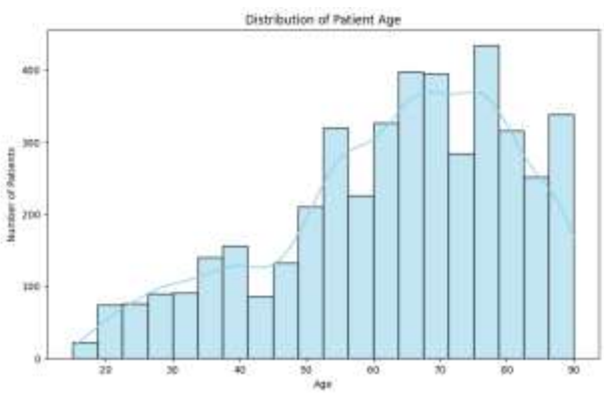

Figure 1.3 Distribution of Patient Age

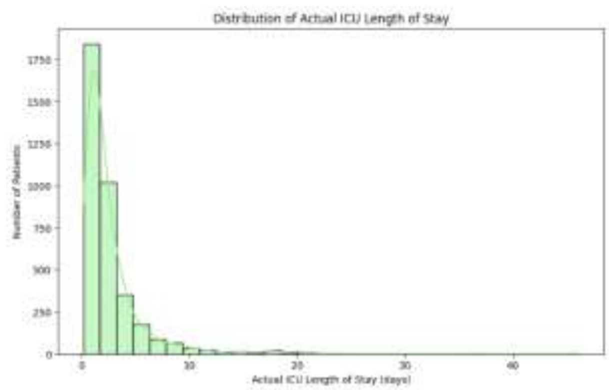

Figure 1.4 Distribution of length of ICU Stay

The age distribution revealed a higher concentration of older patients, with a significant portion above 60 years old. This aligns with expectations in ICU settings and indicates that age is likely an important factor in assessing patient risk.

The distribution of ICU length of stay was right-skewed, with most patients experiencing shorter stays and a smaller subset having prolonged stays. This suggests that while length of stay may be informative, its skewed nature should be considered in modeling..

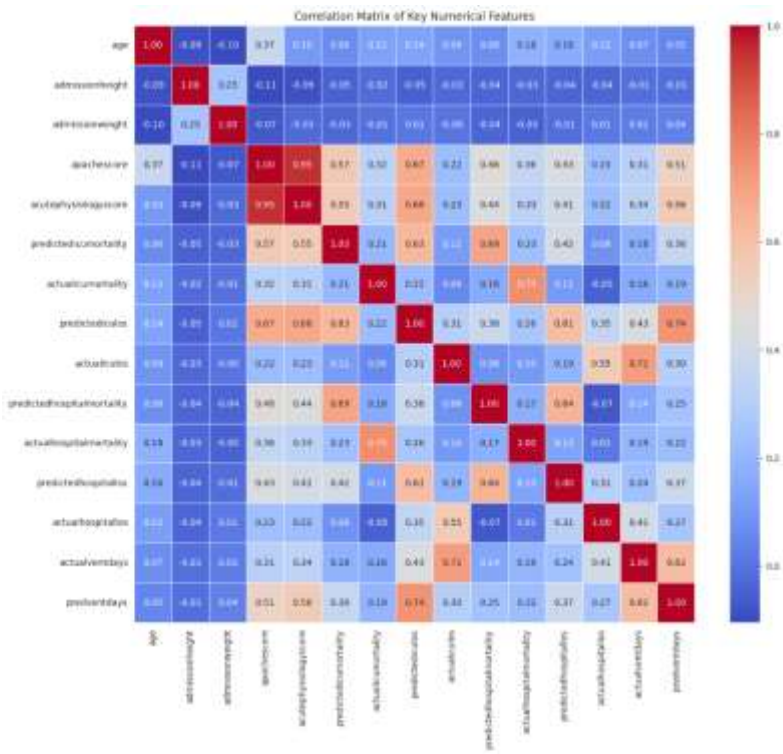

Figure 1.4 Correlation Matrix of Numerical Metrics

The correlation matrix revealed several meaningful relationships between clinical variables and outcomes. As shown in Figure 1.4, APACHE-related scores (apachescore and acutephysiologyscore) showed moderate positive correlations with both predicted and actual ICU and hospital mortality. This confirms their role as strong indicators of patient severity and mortality risk.

Predicted and actual ICU and hospital length of stay were also positively correlated, and both showed some association with mortality. This suggests that longer stays may reflect increased patient severity and higher risk.

Ventilation-related variables (actualventdays and predventdays) were highly correlated with each other and also showed positive relationships with mortality and length of stay, highlighting their connection to more severe clinical conditions.

Lastly, age showed a moderate positive correlation with both APACHE scores and mortality, reinforcing its role as a risk factor. In contrast, admission height and weight showed weaker relationships with outcomes, suggesting they may be less directly predictive.

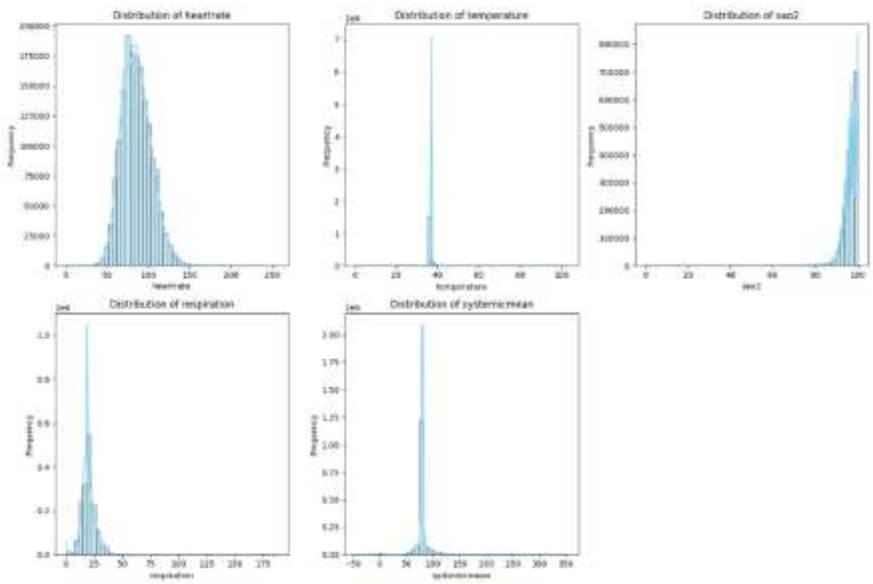

Figure 1.5 Key Vital Signs Distributions

The distributions of key vital signs, including heart rate, temperature, oxygen saturation (SpO_2), respiration rate, and systemic mean blood pressure, all fell within expected physiological ranges but also included extreme values. These extremes likely reflect critical clinical events and highlight the importance of handling outliers carefully.

Figure 1.7 APACHE score by Hospital Mortality

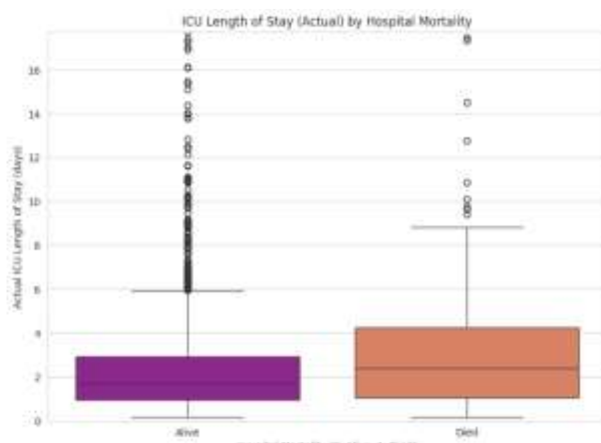

Figure 1.8 ICU LOS by APACHE score

The distribution of APACHE scores differed noticeably between patients who survived and those who did not. Patients who died consistently had higher APACHE scores, confirming that these scores are strong discriminators of mortality risk.

In contrast, ICU length of stay showed a less pronounced difference between outcome groups. While patients who died tended to have slightly longer or more variable stays, the separation was not as strong as with APACHE scores. This suggests that length of stay is informative but less directly predictive of mortality.

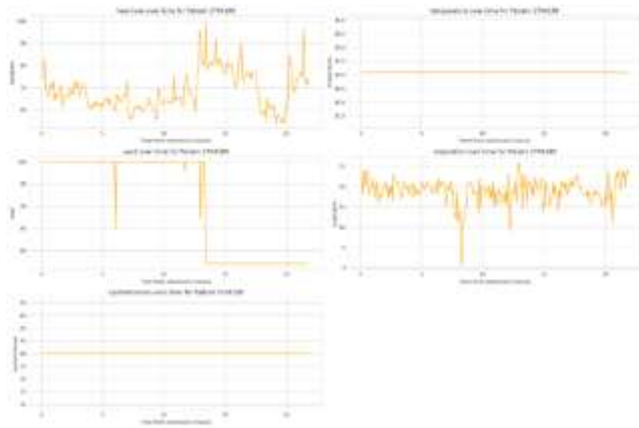

Figure 1.6 Sample Time Series

In addition to the EDA, a time-series plot of a sample patient was constructed to visualize variability in vital signs over time, further emphasizing the dynamic nature of ICU data. As shown in Figure 1.6, this supports the methodological decision to use aggregated statistics rather than single-point measurements in order to better capture patient condition over time.

PCA

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset after feature engineering and encoding. From the original 884 features, 427 principal components were required to retain 95% of the total variance. This represents a substantial reduction in dimensionality while still preserving most of the information in the data..

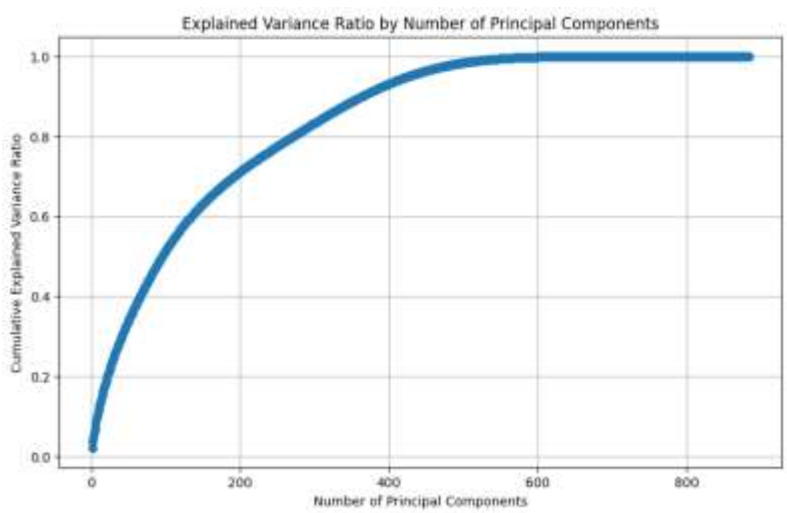

Figure 2.1 PCA Components and Variance

The principal components derived from PCA captured meaningful patterns across clinical variables, even though they do not directly correspond to single interpretable features. Based on the component loadings, several key components can be understood in terms of their underlying clinical relevance.

PC1 reflected overall physiological health and disease severity. It increased with stable lab indicators such as hemoglobin, hematocrit, red blood cell count, and albumin levels, and decreased with higher APACHE and acute physiological scores. This suggests that higher values of this component are associated with better overall health and lower disease severity.

PC4 captured aspects of respiratory function and blood gas exchange. It was influenced by arterial carbon dioxide ($paCO_2$) levels along with red blood cell–related features, indicating its connection to both respiratory status and the blood's ability to transport oxygen and carbon dioxide.

PC2 and PC3 were associated with organ function and physiological balance. PC_2 primarily reflected renal function and electrolyte balance, driven by variables such as blood urea nitrogen and creatinine. PC_3 was linked to inflammatory response and acid-base status, with contributions from white blood cell count, lactate levels, and bicarbonate.

PC119 showed a more complex relationship. It incorporated respiration variability, CO_2-related measures, and discharge-related variables, suggesting that it captures more nuanced patterns in patient trajectories and outcomes that are not as directly interpretable as the earlier components.

Supervised Model Performance

Five supervised learning models were evaluated for hospital mortality prediction: Logistic Regression, Random Forest, LightGBM, XGBoost, and K-Nearest Neighbors (KNN). All models were trained on PCA-transformed features and tested using a stratified train-test split to preserve class distribution.

Figure 3.1 Supervised Models Performance Comparison

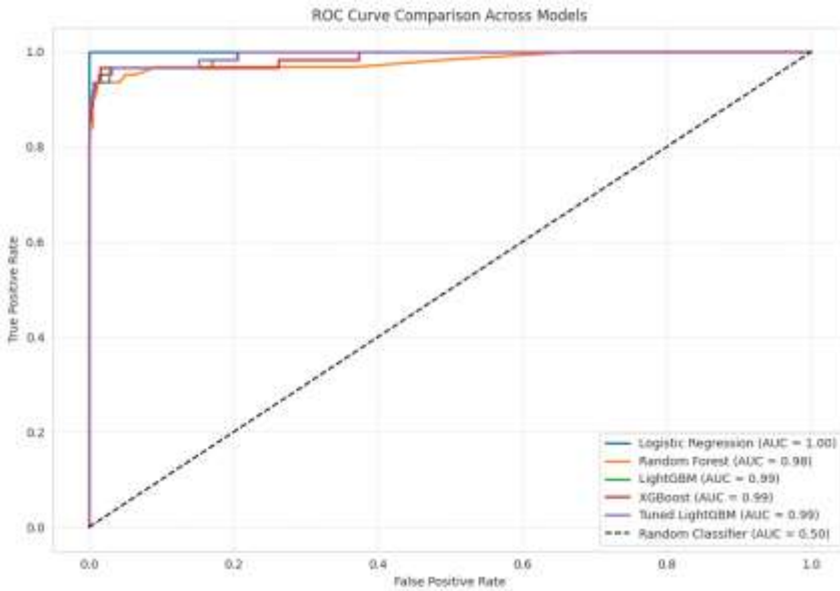

Figure 3.2 Supervised Models ROC Comparison

The overall performance of all models is summarized in the combined heatmap Figure 3.1 which compares Accuracy, Precision, Recall, F1-score, ROC-AUC, and False Positive Rate (FPR) across models. Additional Figure 3.2, helps visualize the ROC Curve comparison across all the models.

Across all metrics, most models performed strongly, reflecting the predictive value of the engineered and reduced feature set. However, clear differences in trade-offs between precision and recall were observed.

Logistic Regression achieved very high recall (1.0) and ROC-AUC (1.0), indicating that it successfully identified all positive cases in the test set. However, this came with a slightly lower precision, suggesting the presence of more false positives compared to some other models.

Random Forest also showed perfect precision (1.0) and zero false positives, but its recall was lower (0.6613), meaning it missed a notable number of actual positive cases. This reflects a more conservative prediction behavior.

LightGBM demonstrated consistently strong performance across all metrics, with high accuracy (0.9920), strong precision (0.9825), balanced recall (0.9032), and a high ROC-AUC score (0.9930). It also maintained a very low false positive rate (0.0012), indicating a strong balance between sensitivity and specificity.

XGBoost performed similarly to LightGBM, with slightly lower precision and ROC-AUC but still strong overall predictive ability across all metrics.

Lastly, KNN performed the weakest among all models, with significantly lower recall (0.2419) and F1-score (0.3571), indicating limited effectiveness in this high-dimensional feature space.

Most Predictive Feature Based on LightGBM

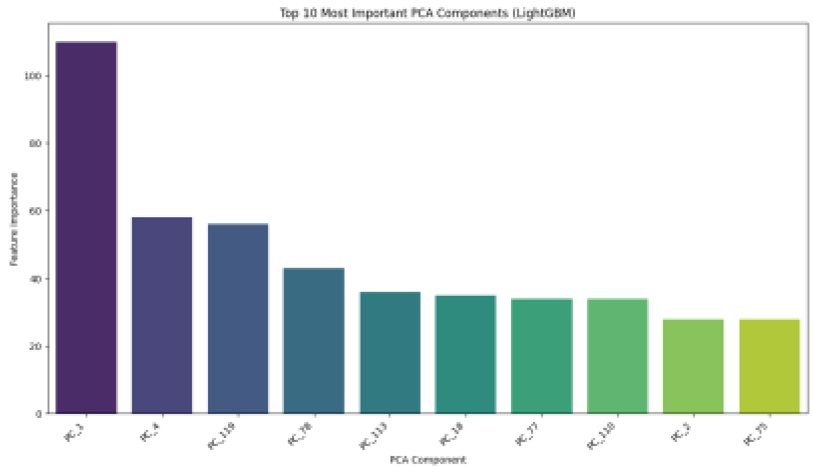

Figure 3.3 Top 10 MOST Important PCA

Figure 3.3 shows the top 10 PCA components ranked by importance in the LightGBM model. PC1 is clearly the most dominant feature by a large margin, indicating that overall physiological health and disease severity are the strongest drivers of mortality prediction. PC4 and PC119 follow, reinforcing the importance of respiratory function and patient trajectory patterns. The remaining components contribute smaller but still meaningful signals

Unsupervised Modelling

Unsupervised learning was applied to explore natural groupings within the patient population based on aggregated vital sign trajectories. The feature set was constructed by summarizing key vital signs from the first 24 hours of ICU admission, including heart rate, temperature, oxygen saturation (SpO_2), respiration rate, and multiple blood pressure measures. These features were expressed using statistical summaries such as mean, median, standard deviation, minimum, and maximum. The resulting dataset was then standardized prior to clustering.

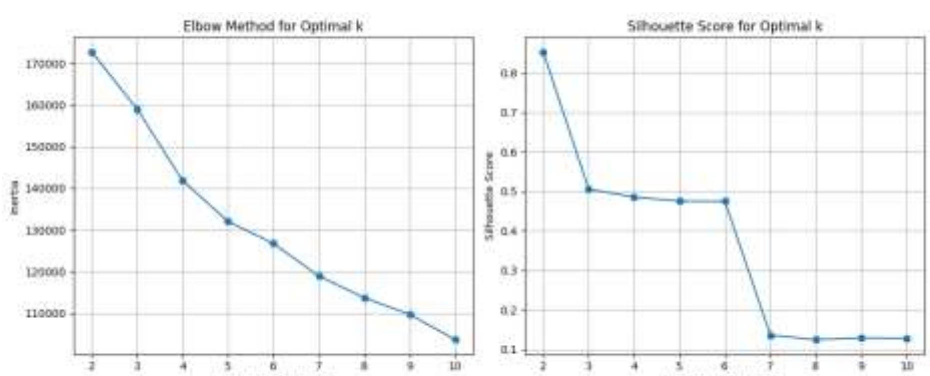

Figure 4.1 K-Means Elbow Method and Silhouette Score

Metric	Cluster 0	Cluster 1
Number of Patients	2367	8
Heart Rate (mean)	83.78	79.27
Temperature (mean)	37.15	37.12
Oxygen Saturation (SpO_2 mean)	95.96	96.92
Respiration Rate (mean)	19.57	21.52
CVP (mean)	16.11	11.72
End-tidal CO_2 (mean)	29.91	30.00
Systemic Systolic BP (mean)	122.73	123.21
Systemic Diastolic BP (mean)	61.39	60.07
Systemic Mean BP (mean)	80.65	79.32
PA Mean Pressure (mean)	25.15	23.84
ICP (mean)	11.03	11.00

Table 4.1 K-Means Cluster Characteristics.

To determine the optimal number of clusters, K-Means was evaluated for values of k from 2 to 10 using inertia and silhouette score. Based on these results, k = 2 was selected as the optimal solution, as it achieved the highest silhouette score (0.85), with a clear decline in clustering quality for higher values of k. This was also supported by the Elbow Method, where inertia gains diminished notably after k = 2. The final clusters are summarized in Table 4.1.

The results show a highly imbalanced distribution, with Cluster 0 containing 2367 patients and Cluster 1 containing only 8 patients.

In terms of clinical characteristics, both clusters show broadly similar vital sign ranges, although Cluster 1 exhibits a slightly higher respiration rate (21.52 vs 19.57) and lower central venous pressure (11.72 vs 16.11). Other measures such as heart rate, temperature, and blood pressure remain largely comparable across clusters.

Overall, K-Means identified one dominant patient group with relatively stable physiological patterns and a very small secondary group with subtle respiratory and hemodynamic differences. The extreme imbalance suggests that the smaller cluster likely reflects rare or outlier patient profiles rather than a distinct and generalizable clinical subgroup.

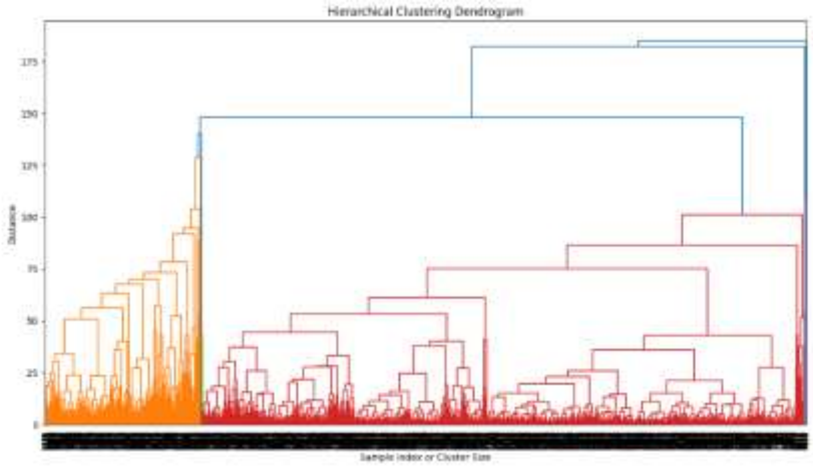

Figure 4.2 Hierarchical Clustering Dendrogram

Metric	**Cluster 1 (N=5)**	**Cluster 2 (N=2369)**	**Cluster 3 (N=1)**
Demographics			
Average Age (Years)	77.0	63.3	73.0
Gender (% Male)	100%	60.5%	100%
Clinical Outcomes			
Hospital Mortality (%)	**20.0%**	8.4%	0.0%
Avg. Ventilation Days	**2.00**	0.97	0.00
Key Vital Signs (Mean)			
Heart Rate (bpm)	80.1	83.8	72.0
Respiration Rate	24.1	19.6	**28.7**
Mean Systemic BP	78.1	80.6	**94.7**
Oxygen Saturation (%)	96.3%	96.0%	96.6%

Table 5.1 Clinical Profiles and Outcomes by Hierarchical Cluster

Feature (Mean)	**Cluster 1**	**Cluster 2**	**Cluster 3**
Systolic BP	122.1	122.7	**136.3**
Diastolic BP	58.7	61.4	**72.8**
CVP (Central Venous Pressure)	12.5	16.1	9.9
PA Mean (Pulmonary Artery)	25.0	25.0	**278.0***
ST Segment 3	**356.2***	0.57	0.02

Table 5.2 Physiological Signature of Clusters

Hierarchical clustering was applied to the aggregated vital sign features, and a 3-cluster solution was selected based on the dendrogram structure and stability of separation.

As shown in Table 5.1, the clustering resulted in a highly imbalanced structure. Cluster 2 contained the majority of patients (n = 2369), while Cluster 1 (n = 5) and Cluster 3 (n = 1) formed very small subgroups.

In terms of outcomes, Cluster 1 had the highest hospital mortality (20%) and the longest ventilation duration (2.0 days), indicating higher severity. Cluster 2 showed lower mortality (8.4%) and shorter ventilation (0.97 days), representing the main clinical population. Cluster 3 had no mortality and no ventilation, suggesting a very low-risk or non-critical cas

Looking at physiological patterns in Table 5.2, Cluster 1 showed lower blood pressure and higher respiration rate compared to Cluster 2, while Cluster 3 showed more extreme and atypical values across several measures, despite being a single observation. Cluster 2 remained the most stable overall.

Overall, hierarchical clustering separated patients into a dominant baseline group, a small high-risk group, and a few extreme edge cases, showing clearer clinical stratification compared to K-Means

Discussion

This study brings together exploratory analysis, dimensionality reduction, supervised learning, and unsupervised clustering to understand and predict hospital mortality in ICU patients. The main idea throughout was not just to build a predictive model, but to understand what the data is actually saying about patient severity, how different physiological systems interact, and whether there are natural patterns in how ICU patients group together. Across all stages, a consistent picture emerges: mortality risk is not driven by a single variable, but by a combination of respiratory, cardiovascular, renal, and general physiological stability signals that interact in complex ways.

The EDA provided the first view of the dataset and confirmed several expected clinical patterns. The ICU population was skewed toward older patients, which aligns with clinical reality. ICU length of stay was heavily right-skewed, meaning most patients had short stays, while a smaller subset had much longer and more complex admissions. Mortality differences across gender and ethnicity were not strongly distinct in a predictive sense, even though raw distributions varied, reinforcing that demographic variables alone are not sufficient to explain outcomes. In contrast, severity-based measures such as APACHE scores showed much clearer separation, with higher scores consistently associated with mortality. This reinforces APACHE as a strong baseline indicator of patient risk. Similarly, correlations between ventilation days, length of stay, and mortality suggest that these variables are not independent but are instead different reflections of overall illness severity.

Vital sign distributions further supported this interpretation. Heart rate, respiration, temperature, oxygen saturation, and blood pressure all fell within expected physiological ranges but included extreme values that likely correspond to critical clinical deterioration. These extremes highlight why simple averages are not sufficient to capture patient condition, and why more structured aggregation and modeling are necessary. The relationship between APACHE scores and outcomes also made this clearer, as patients who died consistently had higher APACHE scores, confirming that severity scoring is strongly tied to mortality even before applying any machine learning models.

When moving into PCA, the goal was to reduce the high dimensionality of the dataset while preserving as much information as possible. Reducing 884 engineered features down to 427 principal components while retaining 95% of the variance shows that the dataset contains a large amount of redundancy but also strong underlying structure. Importantly, even after compression, these components still carry meaningful clinical information.

PC1 stood out as representing overall physiological health and disease severity. It was positively associated with stable lab indicators like hemoglobin, hematocrit, RBC count, and albumin, while showing an inverse relationship with APACHE-related scores. This initially appears counterintuitive, since higher APACHE scores are usually associated with worse outcomes. However, this suggests that PC1 is capturing a broader health dimension that includes underlying risk factors not fully captured by APACHE alone, such as anemia or metabolic imbalance.

PC4 captured respiratory function and blood gas exchange, mainly driven by $paCO_2$ and related hematological measures, reinforcing the importance of respiratory status in ICU outcomes. PC2 and PC3 captured renal function, electrolyte balance, inflammation, and acid-base status, reflecting multi-organ dysfunction across several physiological systems. PC119 was more complex and less directly interpretable, but appeared to capture variability in respiration and discharge-related patterns, suggesting it may reflect patient trajectory or instability over time rather than a single physiological state.

The supervised learning results showed how well these patterns translate into prediction. All models performed strongly overall, confirming that the feature engineering and PCA transformation preserved meaningful predictive signal. LightGBM performed best in terms of overall balance, achieving high accuracy, strong precision and recall, and excellent ROC-AUC while maintaining a very low false positive rate.

Logistic Regression stood out because it achieved perfect recall, meaning it identified all mortality cases, but this came at the cost of lower precision, resulting in more false positives. Random Forest showed the opposite behavior, with perfect precision but lower recall, meaning it was more conservative and missed some true positive cases. XGBoost performed similarly to LightGBM but slightly less consistently across metrics. KNN performed the weakest, which is expected in a high-dimensional PCA space where distance-based methods lose effectiveness.

More importantly than raw performance is what the models are actually learning. Across models, the most influential features were consistently PC1, PC4, and PC119. This alignment is important because it shows consistency between PCA interpretation and predictive importance. In other words, the models are not relying on random or unstable patterns, but repeatedly using the same underlying clinical dimensions: overall physiological severity, respiratory function, and patient trajectory or instability. This strengthens confidence that the PCA representation is clinically meaningful.

The unsupervised learning results provide a complementary perspective by removing the outcome variable entirely and asking whether the data naturally forms meaningful groups. K-Means clustering showed that the optimal structure was very simple, with k = 2 giving the best silhouette score and a clear elbow point. This resulted in one large cluster and one very small cluster. The large cluster represents the dominant ICU population with relatively stable physiological patterns and lower mortality. The smaller cluster shows slightly higher respiratory demand and lower central venous pressure, but its extremely small size suggests it likely captures rare or edge cases rather than a distinct clinical subgroup.

Hierarchical clustering added more structure by producing three clusters instead of two. Here, the largest cluster again represents the baseline ICU population with moderate mortality and stable vital signs. A second cluster clearly represents a higher-risk group, with the highest mortality and longer ventilation duration, indicating more severe illness. A third cluster appears to capture extreme or unusual cases with very low intervention or atypical physiological patterns. Compared to K-Means, hierarchical clustering provides a more nuanced separation of severity levels, showing a clearer gradient from low-risk to high-risk patients.

When comparing supervised and unsupervised approaches, the key takeaway is that both methods converge on the same clinical story, just from different directions. The supervised model directly learns to predict mortality and identifies PC1, PC4, and PC119 as the most important predictive dimensions. The unsupervised models independently discover that patients naturally group based on similar physiological patterns, especially involving respiration, oxygenation, and blood pressure. Even though one approach is predictive and the other exploratory, both highlight the same core systems: respiratory function, cardiovascular stability, and overall physiological balance.

At the same time, they differ in how this information is expressed. Supervised learning provides a direct risk prediction that can be used for decision-making at the individual patient level. Unsupervised learning provides a population-level view, showing that ICU patients are not a single homogeneous group but instead form distinct subgroups with different severity profiles and outcomes.

Overall, the main conclusion is that ICU mortality is inherently multidimensional. It cannot be captured by a single score, a single vital sign, or even a single model. Instead, it emerges from the interaction of multiple physiological systems, and this is consistently reflected across EDA, PCA, supervised modeling, and clustering. What makes the results strong is not just that the models perform well, but that different methods independently converge on the same underlying structure of risk.

Limitations

This project has several limitations. The eICU-CRD dataset is only a subset of the full database, so results may not fully generalize to all ICU populations. Some exploratory analyses used different patient samples (random, stratified, and hospital-based), but the final model was trained on the full merged dataset after PCA.

Feature engineering was also relatively simple, using summary statistics over fixed time windows, which may miss richer temporal dynamics. In addition, while PCA improves performance and reduces dimensionality, it reduces interpretability, making it more difficult to directly link predictions to specific clinical variables.

Finally, the model was only validated internally, so external validation would be necessary before any real-world clinical application.

Future Directions

For future directions, the first priority would be to ensure that the preprocessing pipeline remains consistent when applying the model to new patient data, including cleaning, feature aggregation, merging datasets, and applying the same imputation, encoding, scaling, and PCA steps used during training.

Beyond that, a useful next step would be to improve feature engineering for time-series data by capturing trends over time rather than relying mainly on summary statistics. Another important extension would be to incorporate model interpretability methods such as SHAP to better understand which clinical factors are driving predictions at both the population and individual level.

It would also be valuable to validate the model on external ICU datasets to test how well it generalizes across different hospitals and patient population

Conclusion

Overall, this project shows that ICU mortality can be reasonably predicted using structured clinical data combined with feature engineering and machine learning. The LightGBM model performed strongly after preprocessing and PCA, suggesting that meaningful patterns exist even in high-dimensional ICU data. However, limitations in generalizability, interpretability, and temporal modeling highlight the need for further refinement and external validation before real-world use

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